

Solutions Preparation Sheets for Potency Test: Glucose Stimulated Insulin Release Assay

<https://www.protocols.io/edit/potency-test-glucose-stimulated-insulin-release-as-bc3siyne>

1. Ciprofloxacin Suspension: (Follow instructions in Step 2)

Note: The expiration date is indicated on the Certificate of Analysis and/or bottle. The diluted solution is good for 1 year frozen (if less than CoA expiration date) and 1 month thawed.

Material	Source	Lot#	Expiration Date	Quantity Required	Quantity Used
Ciprofloxacin Powder				500 mg	
Distilled Water				q.s. to 50 ml	

Prepared by _____

Date _____

2. Prepare 1 bottle of PIM(R) media: (Follow instructions in Step 3)

Material	Source	Lot#	Expiration Date	Quantity Required	Quantity Used
PIM (R)				500 mL	
PIM (G) thawed				5 mL	
AB serum (5%v/v)				25 mL	
Ciprofloxacin Solution				0.5 mL	

Prepared by _____

Date _____

3. Krebs Buffer Stock Solution: (Follow instructions in Step 8-14)

Material	Source	Lot#	Expiration Date	Quantity Required	Quantity Used
HEPES powder				5.96 g	
NaCl				6.72 g	
NaHCO ₃				2.02 g	
KCL				0.3728	
MgCl 6H ₂ O				0.2033 g	
BSA				1.0 g	
CaCl ₂ 2 H ₂ O				0.3675g	
Distilled Water				q.s. to 1 L	
1.0 N NaOH				as needed to pH	
1.0 N HCl				as needed to pH	

Prepared by _____

Date _____

4. Stock 280 mM Glucose Solution: (Follow instructions in Step 15-17)

Material	Source	Lot#	Expiration Date	Quantity Required	Quantity Used
D ⁺ Glucose				3.0 g	
Krebs Buffer Stock Solution				60 ml	

Prepared by _____

Date _____

5. High 28 mM Glucose Solution: (Follow instructions in Step 18-19)

Material	Source	Lot#	Expiration Date	Quantity Required	Quantity Used
Stock 280 mM Glucose Solution				5 ml	
Krebs Buffer Stock Solution				45ml	

Prepared by _____

Date _____

6. Low 2.8 mM Glucose Solution: (Follow instructions in Step 20-21)

Material	Source	Lot#	Expiration Date	Quantity Required	Quantity Used
28 mM High Glucose Solution				5 ml	
Krebs Buffer Stock Solution				45ml	

Prepared by _____

Date _____

7. Wash Solution: (Follow instructions in Step 52-54)

Material	Source	Lot#	Expiration Date	Quantity Required	Quantity Used
Concentrated Wash Solution				40mL	mL
Distilled Water				800mL	mL

Prepared by _____

Date _____

8. Anti-Insulin Coagulate Solution: (Follow instructions in Step 55-57)

Material	Source	Lot#	Expiration Date	Quantity Required	Quantity Used
Conjugate Stock Solution				100 µL	µL
Conjugate Buffer				1 mL	mL

Prepared by_____

Date_____

9. Low Insulin Control Solution: (Follow instructions Step 59, 60)

Material	Source	Lot#	Expiration Date	Quantity Required	Quantity Used
Low Insulin Control				1 vial	vial
Distilled Water				500µL	µL

Prepared by_____

Date_____



**Standard Operating Procedure for Potency Test:
Glucose Stimulated Insulin Release Assay**

Details:

Attachment 1: Solutions Preparation Sheets

10. High Insulin Control Solution: (Follow instructions in Step 59, 61)

Material	Source	Lot#	Expiration Date	Quantity Required	Quantity Used
High Insulin Control				1 vial	vial
Distilled Water				500μL	μL

Prepared by_____

Date_____